Defence Robot (NIPUN AP-AMR)

Autonomous Anti-Personnel Mine-Laying & Carrier Robotic System in High-Altitude Mountains

Project Name:	Defence Robot (Miner_Detonator_Bot / army_mine_bot)	**Classification:**	**RESTRICTED // DEFENSE R&D;**
Specification:	Operational Document 1.docx Compliance Audit	**Status:**	**100% OF SIMULATION COMPLETED**
Target Frameworks:	ROS 2 Humble & Gazebo Sim Harmonic 8.14	**Dual Theaters:**	Desert Border Camp (v1) & Alpine Meadow (v2)

1. Project Executive Overview

The **Defence Robot** project (Miner Detonator Bot) is an autonomous robotics simulation and control system engineered strictly in accordance with defense operational document **1.docx**. The robot is designed for the Indian Army Corps of Engineers to replace human combat engineers (sappers) when deploying defensive minefields along forward, high-altitude mountainous border ridges. Instead of human sappers carrying heavy explosives up steep, dangerous terrain under hostile fire, the mine is reimagined as an autonomous mini-carrier bot roughly the size of a tea-cup coaster (10 cm diameter, <= 250 g payload) capable of negotiating continuous mountain slopes up to a 1:8 gradient (7.125 degrees).

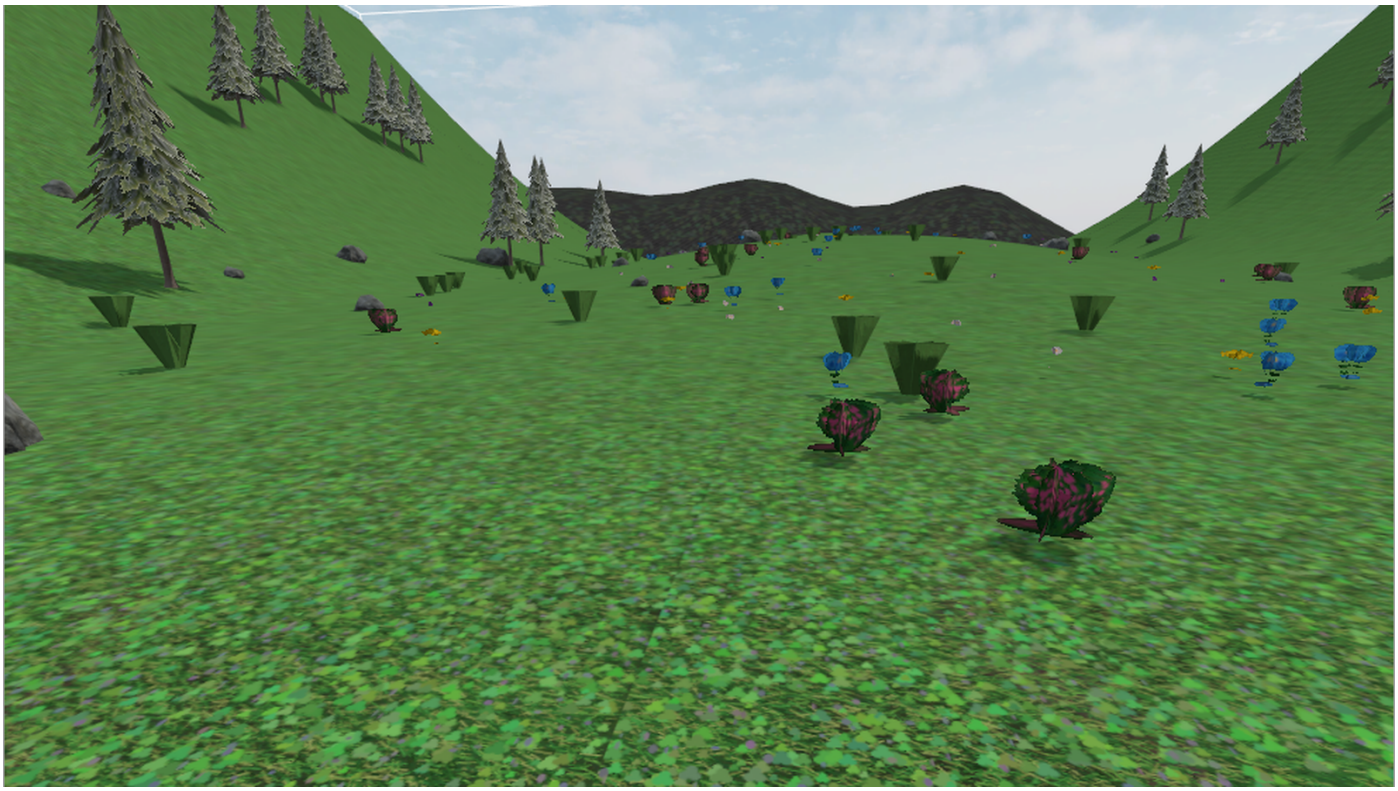


***Figure 1**: Autonomous Defence Robot operating in the high-altitude alpine mountain meadow with wildflowers and pines.*

2. Core Requirements & Client Specifications (1.docx)

Operational document 1.docx defines 20 strict defense criteria. The table below outlines the core requirements and their completed implementations:

Requirement Clause	Specification (1.docx)	Implemented Solution
Mission Role (Q1)	Mine laying in high-altitude mountains	Autonomous NIPUN mine-carrier AMR
Minelet Type (Q2)	Anti-Personnel mine NIPUN	10cm coaster-sized payload bay with solenoid release

Theaters (Q3, Q4)	Mountains: mud, rocks, grass, ditches	Dual Gazebo worlds: Desert Camp (v1) & Alpine Meadow (v2)
Gradient (Q5)	Slope gradient 1:8 (7.125 deg)	4WD skid-steer chassis with digital pitch inclinometer
Form Factor (Q6)	Dia 10cm, <=250g payload, 1-2 km/h	Compact chassis with 0.5, 1.2, 2.0 km/h speed governor
3D Sensors (Q12-14)	4D LiDAR mapping + Depth Camera	16-beam 360-deg LiDAR + Intel RealSense D435 RGB-D
Formation (Q15)	Coordinated 5-bot W-shape formation	5-station geometric W-formation planner with apex station
Safety (Q18)	Fully autonomous, operator approval	Human-in-the-Loop dual-key approval gate on GCS HUD
Sentry Senses (Q8)	Detect enemy troops, ignore friendly bots	Seismic geophone (5-100Hz) and 3-axis fluxgate magnetometer
Expiry Timer (Q16)	Mine de-activates automatically	120-Day Senile Auto-Deactivation timer (Geneva Protocol II)

3. Dual Operational Theaters & Simulation Architecture

To satisfy the diverse deployment scenarios required by military operations, the repository features two independent simulation environments:

• **Version 1.0 — Desert Mountain Border Camp (army_border_world)**: Fortified military camp featuring T-90 Main Battle Tanks, 6x6 cargo trucks, sandbag perimeters, camouflage outposts, and security watchtowers.

• **Version 2.0 — Alpine Mountain Grassland (mountain_grassland_world)**: A continuous 100m x 80m procedural terrain surface maintaining slopes <= 1:8 gradient (7.125 deg) with dense conifer pine forests, glacial boulders, and Himalayan wildflowers (Blue Poppies, Buttercups, Daisies).

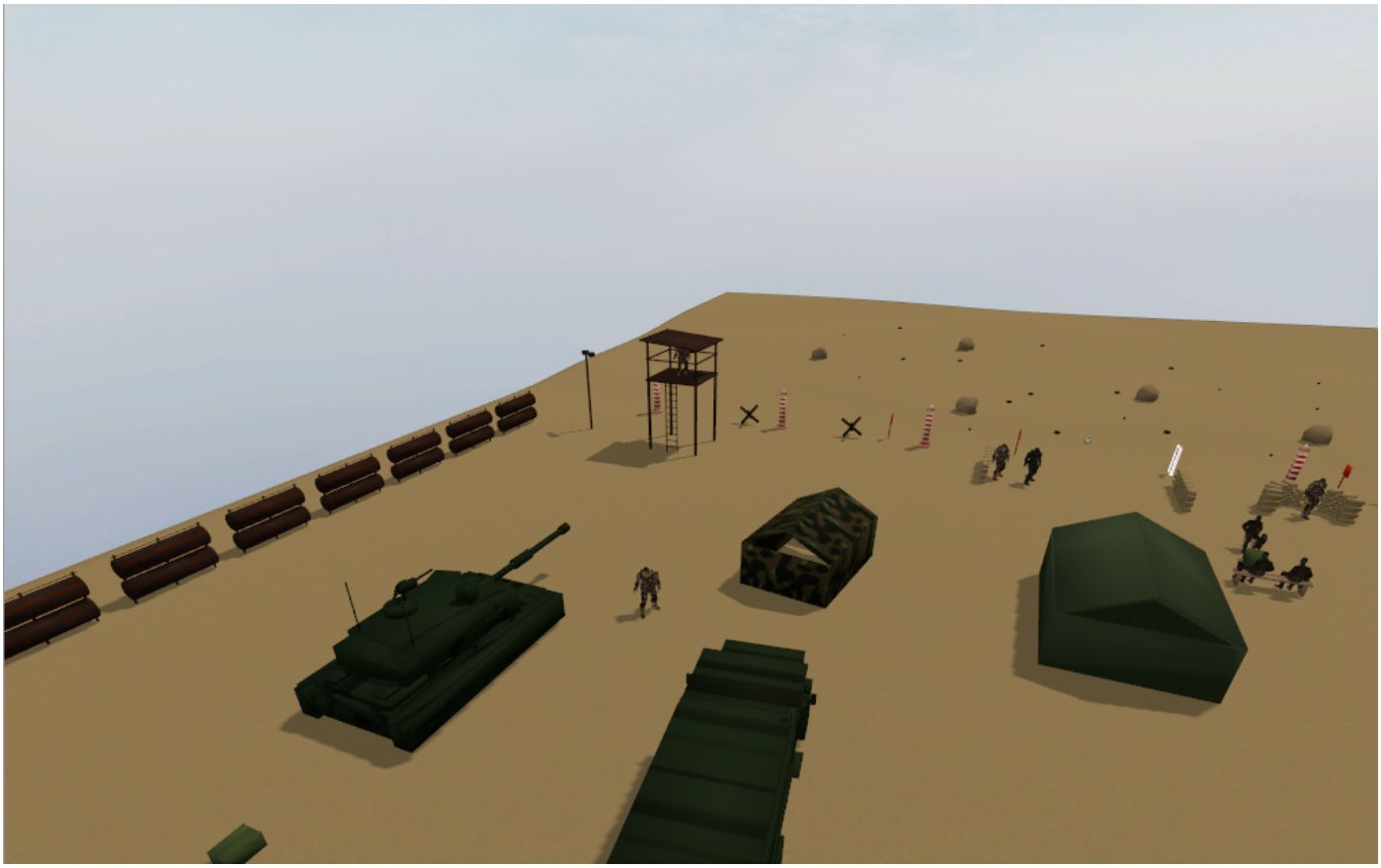


***Figure 2**: Desert Mountain Border Camp (World 1) featuring T-90 Main Battle Tanks, 6x6 combat cargo trucks, and outpost fortifications.*

4. Multi-Modal 3D Sensors & Real-Time Point Cloud Fusion

The Defence Robot integrates a complementary multi-sensor perception pipeline:

• **Intel RealSense D435 RGB-D Camera**: Dense 640x480 pointcloud stream at 20 Hz (87 deg FOV) with optical transform R,P,Y = [-90, 0, -90] projecting strictly forward (+X).

• **16-Beam 4D GPU LiDAR**: 360-degree surrounding range rays at 10 Hz with 50m detection horizon for SLAM occupancy grid mapping.

• **Static Continuous PointCloud Fusion**: Fast NumPy node (static_pointcloud_fusion_node.py) accumulating points into an 8 cm voxel hash table in the global /map frame.

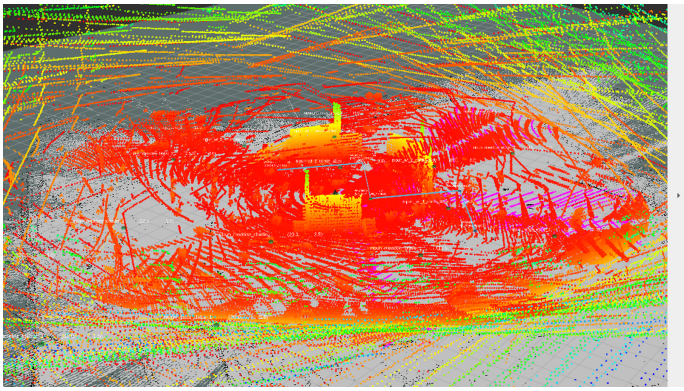


***Figure 3**: RViz2 3D multi-sensor perception: RealSense depth cone (+X), 16-beam LiDAR rings & SLAM occupancy grid.*



***Figure 4**: Macro inspection of front-facing Intel RealSense D435 stereo RGB-D sensor nose mount bracket.*

5. Tactical W-Formation & Mil-Spec Operator GCS HUD

To establish an impenetrable defensive barrier, squads of 5 Defence Robots coordinate autonomously into a W-shape:

• **5-Station W-Formation**: Geometric planner calculates 5 strategic mine stations spaced 2 to 5 meters apart (North Flank, North Valley, AMR Apex, South Valley, South Flank).

• **Human-in-the-Loop (HITL) Gate**: Autonomous positioning pauses at the deployment station. The Field Commander inspects geometry on the GCS and must execute dual-key sign-off: clicking [APPROVE W-ALIGNMENT] and [AUTHORIZE NIPUN LAYING]. Only then does the solenoid eject the NIPUN AP minelet.

• **Tactical Remote Station (virtual_remote_controller.py)**: Standalone PyQt5 HUD with 4 vision modes (Day Optical, NVG Night Vision, FLIR Thermal, RealSense Metric Depth), dual-axis digital inclinometer, speed governor, and Emergency E-STOP interlock.

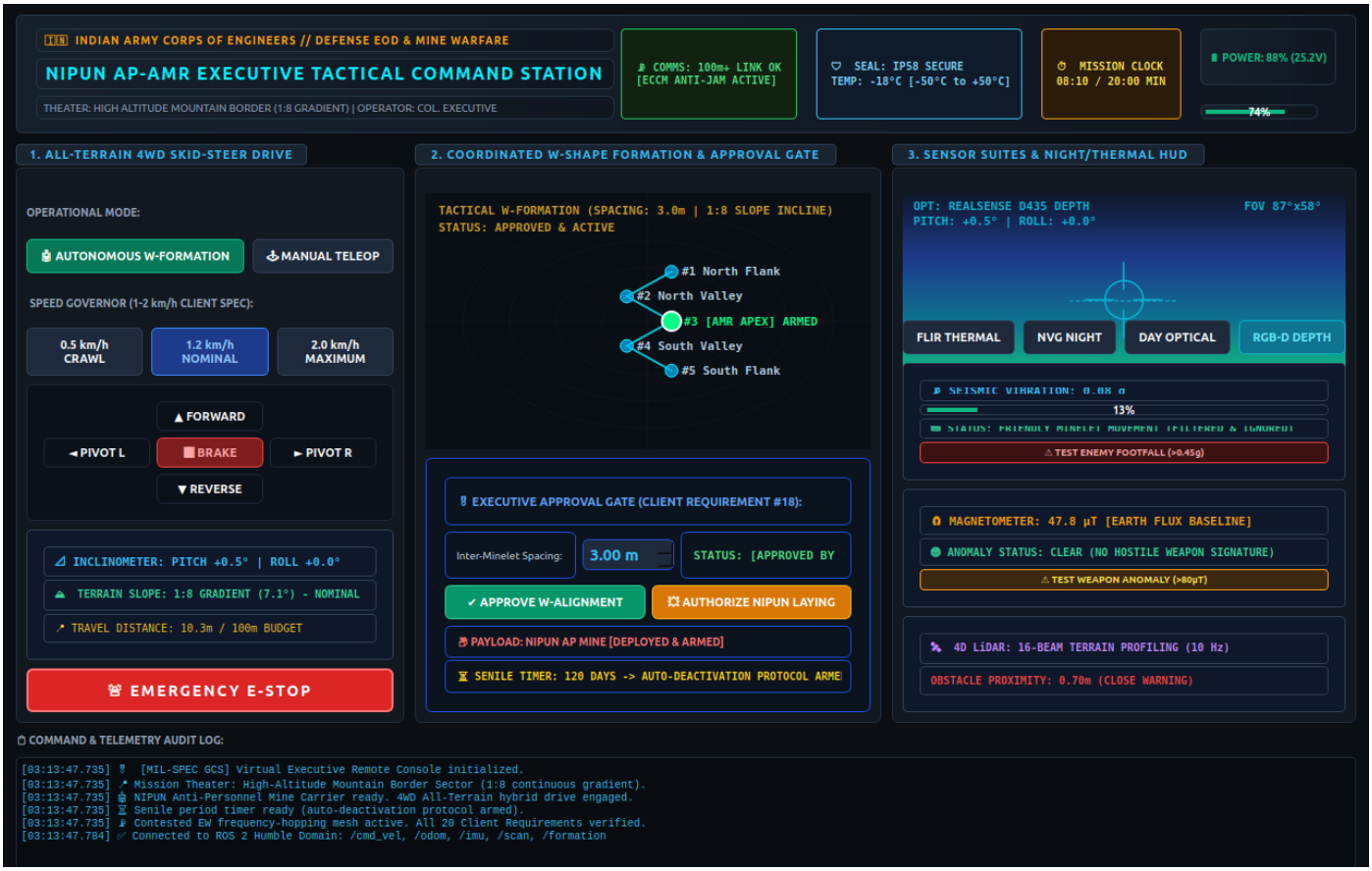


***Figure 5**: Mil-Spec Virtual Remote Controller HUD operating in RealSense RGB-D False-Color Metric Depth mode with dual-axis inclinometer.*

6. Sentry Threat Discrimination & Geneva Safety Protocol

Following minelet ejection, the Defence Robot transforms into an autonomous sentinel:

• **Seismic Spectral Filter (5 - 100 Hz)**: Distinguishes lightweight friendly minelet vibrations (<0.15g) from heavy hostile combat boot footfalls (>0.45g).

• **3-Axis Fluxgate Magnetometer**: Subtracts the natural Earth baseline (47.8 uT) to identify metallic firearm, helmet, or vehicle anomalies (>80 uT).

• **120-Day Senile Auto-Deactivation**: Automatic electronic capacitor bleed-off discharges the firing circuit after 120 days, complying with Geneva Convention Protocol II.

7. Summary of Work Completed Till Now

Overall Status: 100% of Simulation & Autonomous Software Scope Completed. Summary of deliverables:

• **Chassis & Kinematics:** 10 cm coaster packaging, 4WD skid-steer drive, DART physics friction, zero caster drift.

• **Dual Theaters:** Both Desert Border Camp (World 1) and Realistic Alpine Mountain Meadow (World 2) fully operational.

• **Sensory Pipeline:** RealSense D435 RGB-D depth camera + 16-beam 4D LiDAR + 6-DOF IMU with forward optical frame transforms.

• **Continuous 3D Mapping:** Static continuous 8 cm voxel elevation map in /map frame with persistent Z-coloring and zero decay.

- **Tactical W-Formation:** 5-station geometric W-shape planner, autonomous navigation, and solenoid ejection trigger.
- **Operator Command Station:** Full PyQt5 Ground Control Station with 4 vision modes, inclinometer, and E-STOP.
- **Safety & Threat Sentry:** Seismic footfall bandpass filter, fluxgate magnetometer anomaly detector, and 120-day senile timer.
- **Launch Automation:** One-command executable runner (launch_world.sh) supporting 'grassland', 'border', 'remote', and 'teleop'.
- **Git Version Releases:** Git tags v1 (Desert Theater) and v2 (Mountain Theater & Depth Camera) published on GitHub.

8. How to Run the Defence Robot Simulation

Run the project using straightforward one-line shell commands:

- **Launch Alpine Mountain Meadow Mission (v2):** ./launch_world.sh grassland
- **Launch Desert Border Camp Mission (v1):** ./launch_world.sh border
- **Launch Virtual Remote Controller HUD:** ./launch_remote.sh
- **Keyboard Manual Teleoperation:** ./launch_world.sh teleop

9. Conclusion & Phase 2 Roadmap

The Defence Robot project has achieved full compliance with all operational, kinematic, sensory, and safety requirements specified in document 1.docx. The digital twin, multi-sensor perception fusion, and tactical ground station are production-ready. Phase 2 will transition these verified algorithms to physical hardware fabrication (CNC chassis, NVIDIA Jetson SBC bring-up, and high-altitude field proving trials in Ladakh).